MIL-C-3098/123A (USAF)
7 April 1975
SUPERSEDING
MIL-C-3098/123 (USAF)
2 February 1970

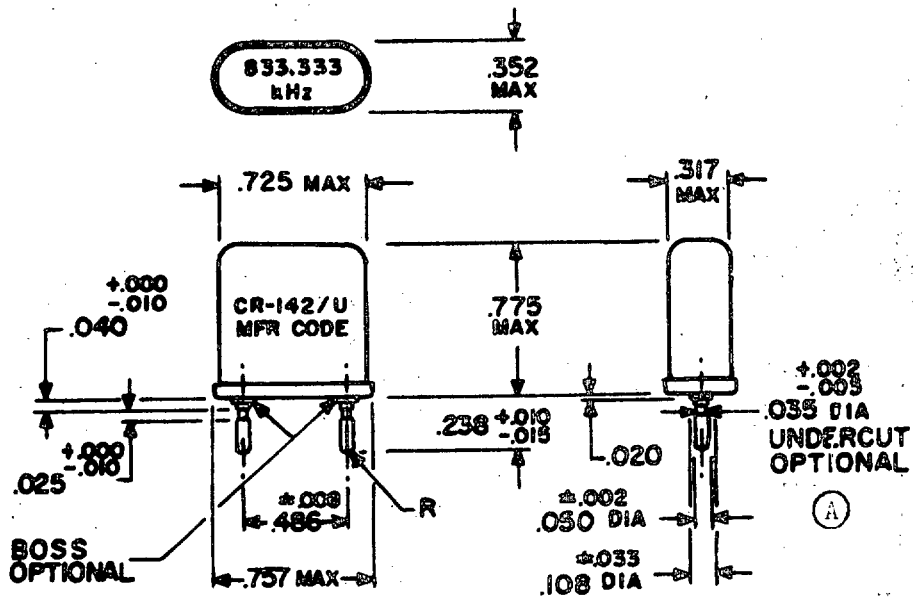
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-142/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the crystal unit, quartz, described herein shall consist of this document and the latest issue of MIL-C-3098.

Pertinent characteristics: 833.333 kHz; fundamental mode; controlled; antiresonance.



INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.002	.05	.020	.51	.050	1.27	.486	12.34
.005	.13	.025	.64	.108	2.74	.725	18.42
.008	.20	.033	.84	.238	6.05	.757	19.23
.010	.25	.035	.89	.317	8.05	.775	19.69
.015	.38	.040	1.02	.352	8.94		

NOTES:
1. DIMENSIONS ARE IN INCHES.
2. METRIC EQUIVALENTS (TO THE NEAREST .01 mm) ARE GIVEN FOR GENERAL INFORMATION ONLY AND ARE BASED UPON 1 INCH = 25.4 mm.

(A) Denotes Change

FSC 5955

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TABLE I. REQUIREMENTS

Requirement	Specified Value or Condition
Holder	HC-6/U (MIL-H-10056/)
(A) Dimensions, marking, and configuration	See figure
Frequency	833.333 kHz
Capacitance, shunt	7.0 pf, maximum
Mode of oscillation	Fundamental
Government reference standard test set	TS-330/TSM
Rated drive level	5.0 ±1.0 mw 1/
Calibration values	
Equivalent-resistance	Zero ohms
Crystal current	Note 2/
Antiresonance, load capacitance	32.0 ±0.2 pf
Reference temperature	75° ±1°C
Operating temperature range (controlled)	70° to 80°C, inclusive
Frequency tolerance	±0.001 percent
Equivalent resistance	275 ohms, maximum
Frequency stability	±0.0005 percent
Operable Temperature Range	-55° to +70°C and 80° to 90°C, inclusive
Shock:	
Frequency change permitted	±0.0005 percent
Equivalent resistance	±15 percent
Vibration: 3/	
Frequency change permitted	±0.0005 percent
Equivalent resistance change permitted	±15 percent
Temperature cycling:	
Frequency change permitted	±10 hertz
Equivalent resistance change permitted	±20 percent
Aging:	
Frequency change permitted	0.0005 percent

1/ The violation of $P = I^2R$ is intentional; at the higher resistance of the crystal unit, the empirical power dissipation will be as rated.

2/ Set crystal current control at extreme counterclockwise (minimum). position.

3/ Method 201 of MIL-STD-202.

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NOTES:

A. These crystal units are for use in the Magnavox Company MX-1489/ARC-27 (1st Oscillator) equipment. They were previously fabricated, using Air Force Technical Order 12-1-20 or Magnavox Drawing No. 537912.

Custodians:
Air Force - 84

Review activities:
Air Force - 11, 85

User activities:
Air Force - 17

Preparing activity:
Air Force - 84

(Project No. 5955-F171-16)

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